Fiscal Year:	FY 2022	**Task Last Updated:**	FY 04/11/2022
PI Name:	Quintiere, James Ph.D.		
Project Title:	Experimental Investigation of Emulated Burning Rate at Various Gravity Levels		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20742-3031	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2015	**End Date:**	02/08/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 02/08/2023 per NSSC information (Ed., 4/11/22) NOTE: Extended to 02/08/2022 per NSSC information (Ed., 12/3/21) NOTE: New grant awarded 80NSSC20M0119 with new start/end dates 4/9/2020-10/08/2021; added on to existing grant NNX15AD06A per D. Stocker/GRC (Ed., 10/13/20) NOTE: Extended to 12/31/2020 per NSSC information (Ed., 11/7/19) NOTE: Extended to 12/31/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) deRis, John Ph.D. (FM Global)		
Grant/Contract No.:	80NSSC20M0119 ; NNX15AD06A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: New grant awarded 80NSSC20M0119 with new start/end dates 4/9/2020-10/08/2021; added on to existing grant NNX15AD06A per D. Stocker/GRC (Ed., 10/13/20) NOTE: Continuation of "Experimental Investigation of Emulated Burning Rate at Various Gravity Levels," grant NNX10AD98G, with the same Principal Investigator (PI) (Ed., 1/14/2015) A novel experimental apparatus, the Burning Rate Emulator (BRE), is being developed. This apparatus is capable of assessing ignitability, steady burning rate, and extinction characteristics of 'virtual' solid and liquid fuels as a function of gravity level, flow characteristics, and orientation. By changing the fuel gas and its flow rate, it will be possible to emulate steady combustion for a range of liquid and solid fuels having B numbers (mass transfer numbers) and radiation characteristics that fall anywhere within physically realistic bounds. Measurements of the critical mass flux at ignition can also be made. These measurements will be conducted as a function of scale and orientation to assess turbulent and laminar burning in natural convection. The experiment is well suited to testing in the Combustion Integrated Rack aboard the International Space Station (ISS). The Burning Rate Emulator (BRE) will simulate the steady burning of liquids and solids using a rectangular porous burner fed by gaseous fuel and equipped with a total heat flux sensor. BRE will control the fuel flow rate based on the measured heat flux at the burner and the measured surface temperature. This will allow it to use a small number of gaseous fuels to match the following four properties of a large number of condensed fuels: heat of combustion (LHV), heat of vaporization (L), surface temperature (Ts), and laminar smoke point. Extinguishment will be considered via changes in fuel flow rate. Both low flow rates (for kinetic extinction) and high flow rates (for radiative extinction) will be considered. The BRE flight program seeks to establish the burning conditions in a quiescent microgravity ambient in terms of material properties. The specific flight objectives are as follows: 1. Observe burning behavior in long-duration microgravity for CH4/N2 and C2H4/N2 mixtures on porous circular burners of 25 and 50 mm diameter. Consider O2/N2 ambients with various pressures and ambient O2 concentrations and pressures. NASA-considered atmosphere for human space missions will be examined: 14.7 psia, 21% oxygen: 10.2 psia, 26.5%; and 8.2 psia, 34% oxygen. Emphasize fuel conditions that can be related to a diverse range of condensed fuels. 2. Consider the conditions of Objective 1 for ambients of O2/CO2 (desired). 3. Consider the conditions of Objectives 1 – 2 for extinction limits.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is expected to improve the understanding of fire and combustion processes in both microgravity and normal gravity. It also could advance the state of the art in the design of porous burners and simple heat flux meters.
Task Progress:	We completed the Round 2 testing aboard the International Space Station (ISS). We continued working on the data analysis of all Burning Rate Emulator (BRE) data. Three journal papers and four conference papers and posters were published this year. We have performed power-law regression analyses of radiation fraction and surface heat flux with test conditions, i.e., mass flux, surface temperature, the heat of combustion, oxygen mole fraction, and pressure. Regression analyses were performed on the ISS measurements to examine correlations of burning rate. The best correlations obtained to date correlate burning rate with oxygen mole fraction and heat of gasification. Pressure has no statistical effect on burning rate. A flammability map is in progress in terms of heat of combustion vs. heat of gasification for each ambient condition. It will identify flammable regions. We continued to analyze the thin-filament pyrometry (TFP) results, and the correlations between radiative loss fraction, adiabatic flame temperature, and peak temperature. We are collaborating with Alex Snegirev to obtain Computational Fluid Dynamics (CFD) runs of these flames. [Ed. Note: See bibliography.] We extracted steady heat flux on the burner surface and flame radiation measurements analysis to report heats of gasification (which corresponds to a range of solids) and radiative loss fraction correlation with flame height. We developed a MATLAB code to read an image sequence with an unlimited number of images and to measure flame dimensions for each image.
Bibliography Type:	Description: (Last Updated: 10/08/2025)
Articles in Peer-reviewed Journals	Snegirev A, Kuznetsov E, Markus E, Dehghani P, Sunderland PB. "Transient dynamics of radiative extinction in low-momentum microgravity diffusion flames." Proceedings of the Combustion Institute. 2021;38(3):4815-23. https://doi.org/10.1016/j.proci.2020.06.110 , Jan-2021
Articles in Peer-reviewed Journals	Dehghani P, Sunderland PB, Quintiere JG, deRis JL. "Burning in microgravity: Experimental results and analysis." Combustion and Flame. 2021;228:315-330. https://doi.org/10.1016/j.combustflame.2021.01.035 , Jan-2021
Articles in Peer-reviewed Journals	Dehghani P, Quintiere JG. "Theoretical analysis and predictions of burning in microgravity using a burning emulator." Combustion and Flame. 2021 Nov;233:111572. https://doi.org/10.1016/j.combustflame.2021.111572 , Nov-2021